

Constructing the epigenetic clock of fish using the Atlantic herring as an example – preliminary results.

Can DNA methylation support fish age estimation?

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DNA methylation age predictors have already been tested in fish

Examples:

- **Zebrafish (*Danio rerio*)** – Mayne B. et al. **2020**, A DNA methylation age predictor for zebrafish. *Aging*, 12(24), 24817–24835
- **European seabass (*Dicentrarchus labrax*)** - Anastasiadi D. and Piferrer F. **2020**, A clockwork fish: Age prediction using DNA methylation-based biomarkers in the European seabass. *Mol. Ecol. Resour.* 20: 387–397
- **Medaka (*Oryzias latipes*)** – Bertucci E.M. et al. **2021**, The aging DNA methylome reveals environment-by-aging interactions in a model teleost. <https://doi.org/10.1101/2021.03.01.433371>: bioRxiv preprint
- **Red snapper (*Lutjanus campechanus*), Red grouper (*Epinephelus morio*)** – Weber D.N. et al. **2022**, Novel epigenetic age estimation in wild-caught Gulf of Mexico reef fishes. *Can. J. Fish. Aquat. Sci.* 79: 1–5
- **Blackbelly rosefish (*Helicolenus dactylopterus*)** – Weber D.N. et al. **2024**, Epigenetic age estimation in a long-lived, deepwater scorpionfish: insights into epigenetic clock development. *Can. J. Fish. Aquat. Sci.* 00: 1–12
- **Haddock (*Melanogrammus aeglefinus*)** – Strand E.L. et al. **2025** Building an ‘epigenetic clock’: Utilizing whole genome DNA methylation patterns to predict age in haddock. <https://doi.org/10.1101/2025.10.16.682892>: bioRxiv preprint
- **Atlantic cod (*Gadus morhua*)** – Anastasiadi D. et al. **2026**, An Epigenetic Clock for Accurate Age Prediction in Atlantic Cod Populations for Improved Fisheries Management. *Apr*;26(3):e70109, doi: 10.1111/1755-0998.70109

Why herring?

- important species for fisheries,
- wild material is available,
- individuals cover a useful age range,
- age can be estimated from otoliths,
- different populations and environments can be tested later.



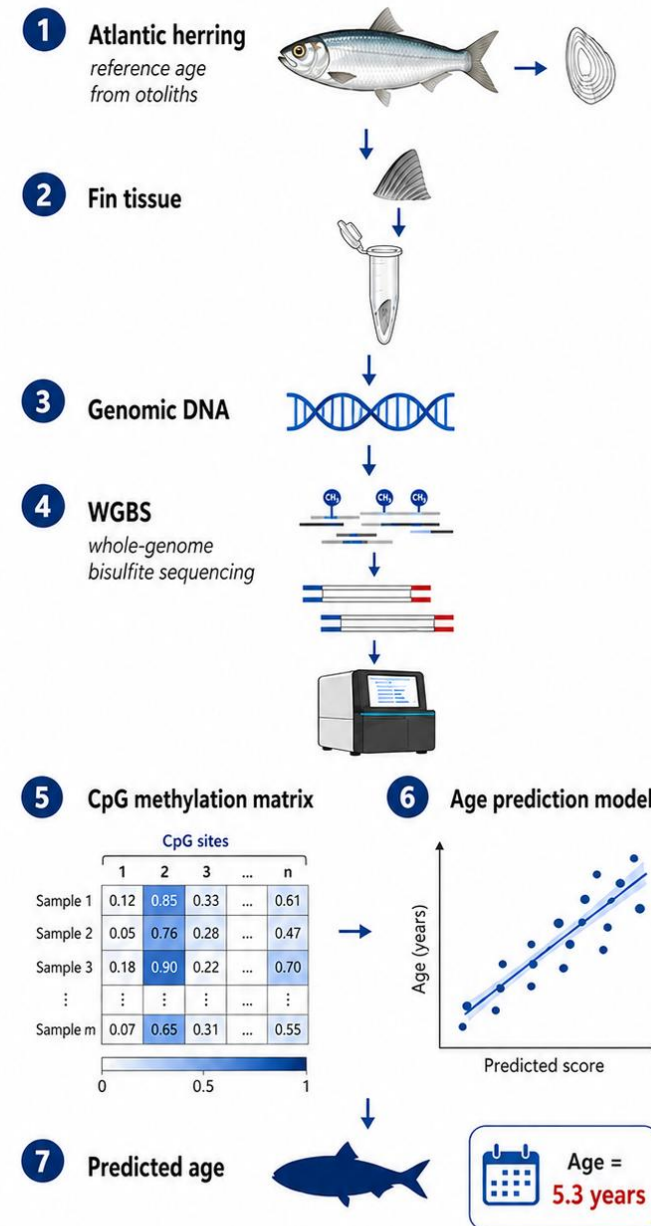
Aim

Testing the workflow of identifying candidate CpG markers associated with age in Atlantic herring

Main steps

1. fish collection and age determination (otoliths)
2. tissue sampling (fin clippings)
3. DNA extraction
4. Whole Genome Bisulfite Sequencing (WGBS) analysis
5. methylation data processing
6. candidate age-related CpG markers selection
7. age prediction model building and validation

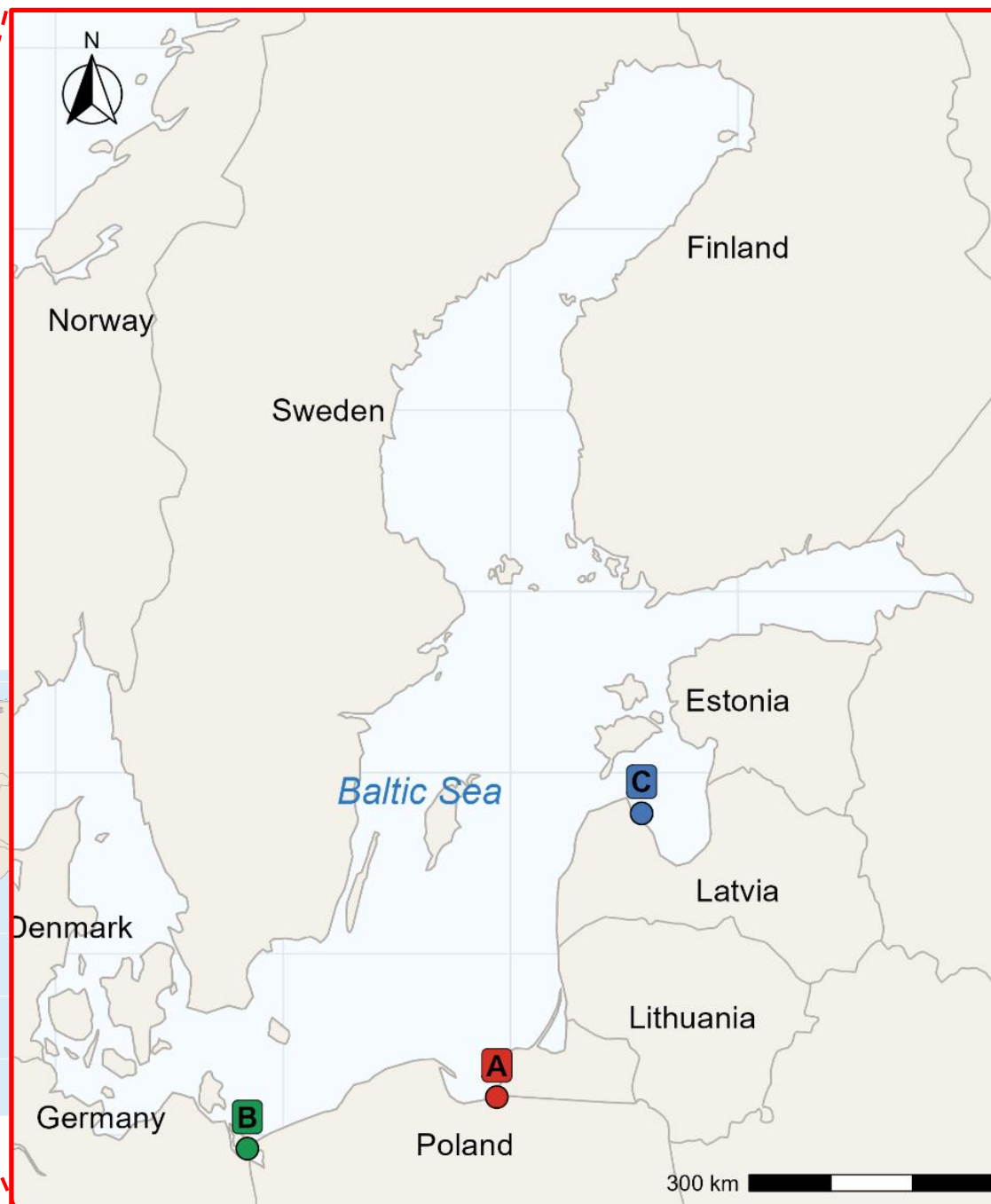
Workflow for DNA methylation-based age prediction



Study material

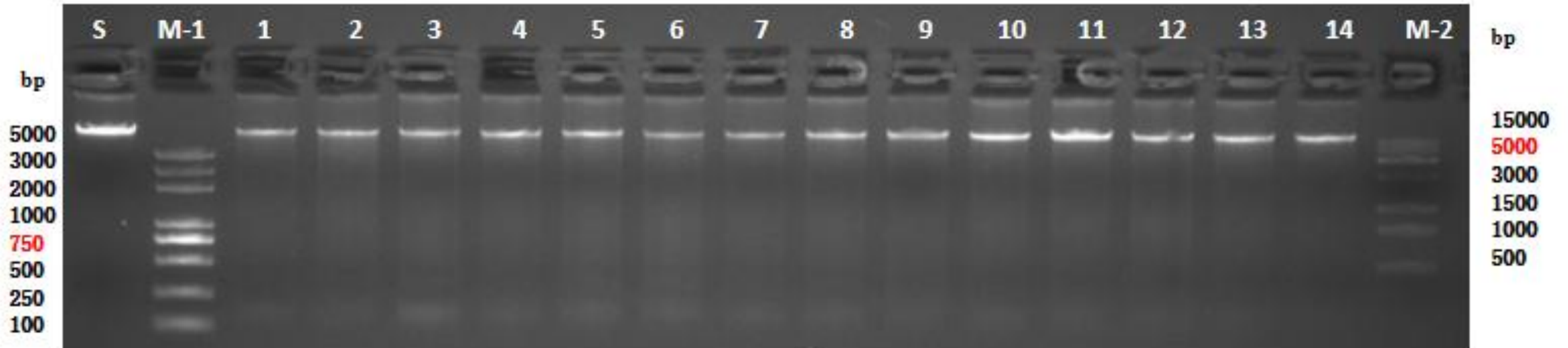
- Current WGBS model: 16 fish from Vistula Lagoon, age rang 3–10 years
- Additional material: younger fish (0-10 years), reference and other population for validation

- A** - Vistula Lagoon
- B** - Pomerania Bay
- C** - Gulf of Ryga



DNA extraction and sample quality

- fin tissue was used for DNA extraction
- DNA was extracted using a column-based kit
- concentration and purity were checked (in 50 ul ranged from 27 – 140 ng/ul)
- selected samples were sent for WGBS (100 ng of genomic DNA per sample is required)



How methylation was measured: WGBS

Main steps:

1. genomic DNA
2. DNA fragmentation
3. lambda DNA spike-in and bisulfite conversion
4. library preparation
5. Illumina sequencing
6. methylation calling

Bisulfite conversion:

methylation protects cytosine during bisulfite treatment

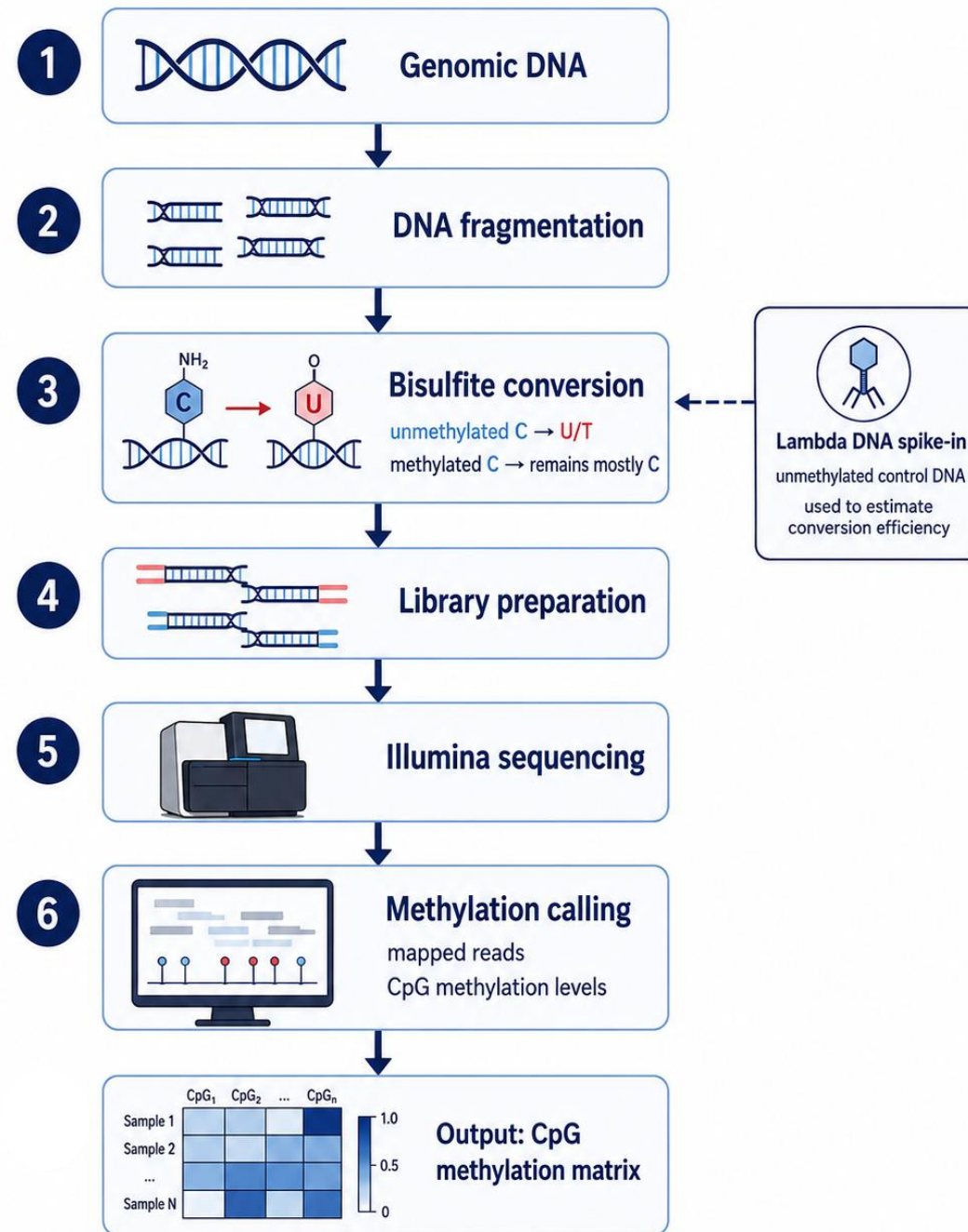
- unmethylated C $\rightarrow$ U/T
- methylated C $\rightarrow$ remains mostly C

Lambda DNA:

unmethylated external control DNA

most lambda cytosines should be converted to U/T

used to estimate conversion efficiency



from genomic DNA to CpG methylation data

From sequencing reads to methylation values- bioinformatic processing

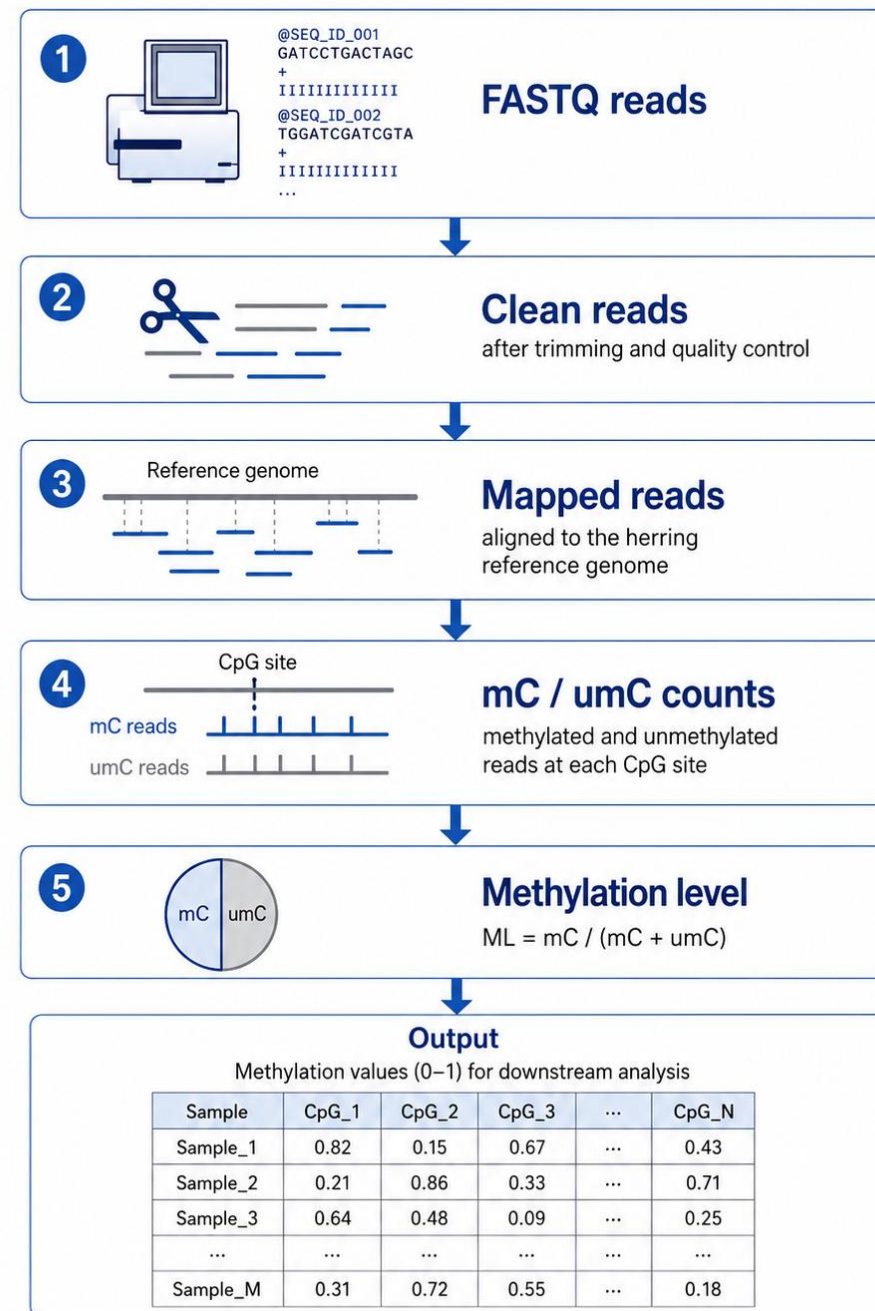
1. FASTQ reads
2. trimming and quality control
3. mapping to the herring reference genome and PCR duplicate removal
4. methylation extraction
5. methylation level:

$$ML = mC / (mC + umC)$$

mC = methylated reads

umC = unmethylated reads

Methylation level ranges from 0 to 1.



The sequencing data passed the main quality checks.

Parameter	Result	Meaning
Q20	~99%	high-quality bases
Q30	~95–96%	very high-quality bases
Clean reads	~85–88%	reads kept after filtering
Conversion efficiency	99.1–99.7%	successful C → U/T conversion
Mapping rate	~70.4–73.4%	reads placed on reference genome
Unique mapping	~66%	reads placed in one genomic location

What does Q30 or Q20 mean?

Q30 means that the expected base-call accuracy is about 99.9%, roughly one error per one thousand bases

Q20 means that the expected base-call accuracy is about 99%, roughly one error per one hundred bases

How to calculate conversion efficiency?

$$\text{conversion} = 1 - \frac{\text{number of unchanged cytosines}}{\text{total number of cytosines}} \%$$

Filtering CpG sites before modelling

We kept CpG sites that were:

in CpG context,

located on nuclear chromosomes,

covered at $\geq 10\times$,

present in all 16 samples,

variable across samples

Final candidate set: 2,919,929 CpG sites

Filtering CpG sites before modelling

Atlantic herring methylation data



CpG context only



nuclear chromosomes only

STEP

1

All CpG positions on nuclear chromosomes

14,869,806 CpG positions

STEP

2

Coverage filter

coverage $\geq 10\times$



higher coverage =
more reliable
methylation
estimates

STEP

3

Common CpG sites

present in all samples

5,986,518 common CpG positions

STEP

4

Variance filter

remove almost constant sites

STEP

5

Candidate CpG predictors for modelling

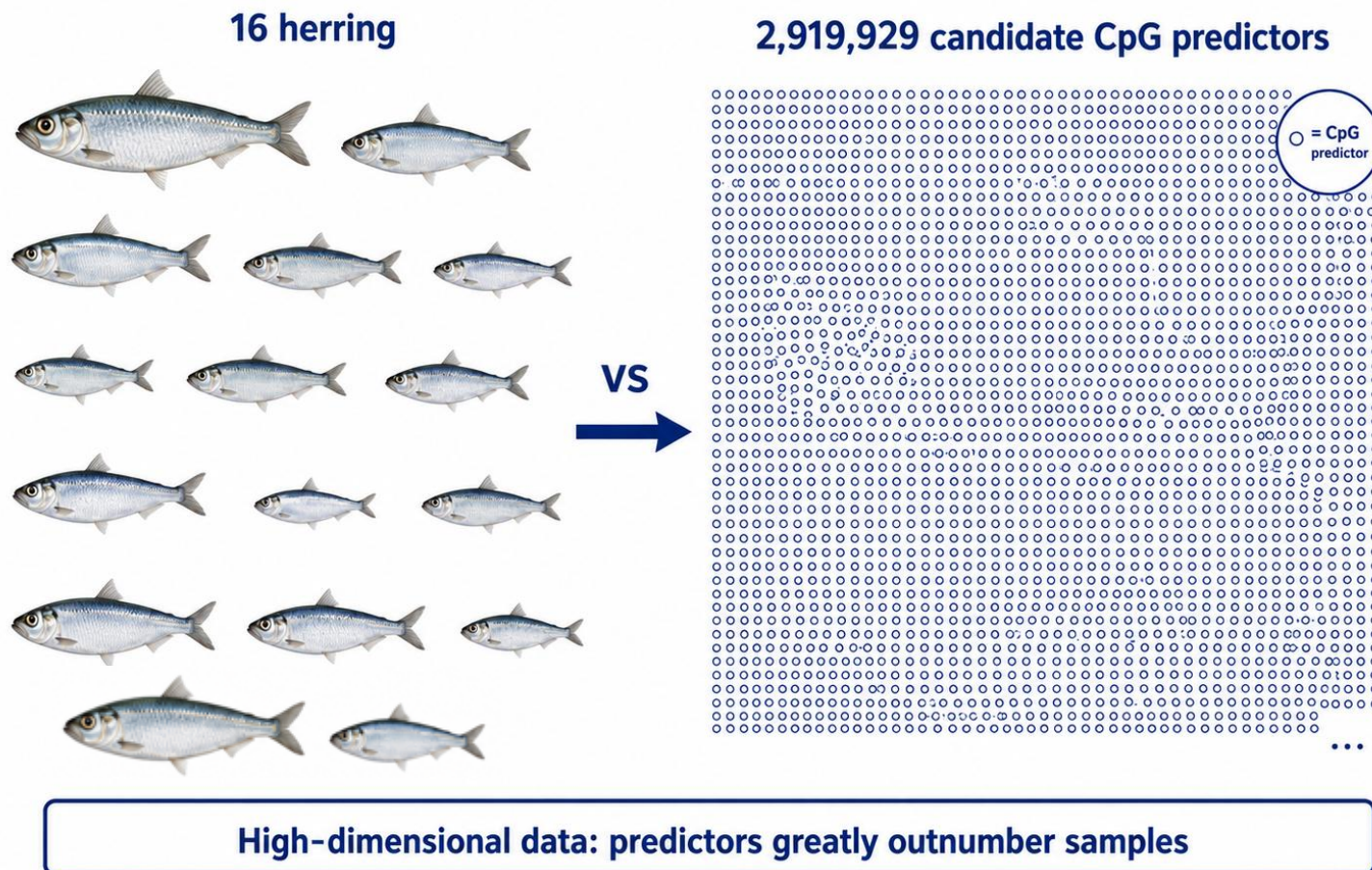
2,919,929 CpG sites

A high-dimensional modelling problem

Input data for modeling:

- 16 herring samples
- 2,919,929 candidate CpG predictors
- response variable: **age**
- predictors greatly outnumber samples

Main risk: overfitting



Why elastic net?

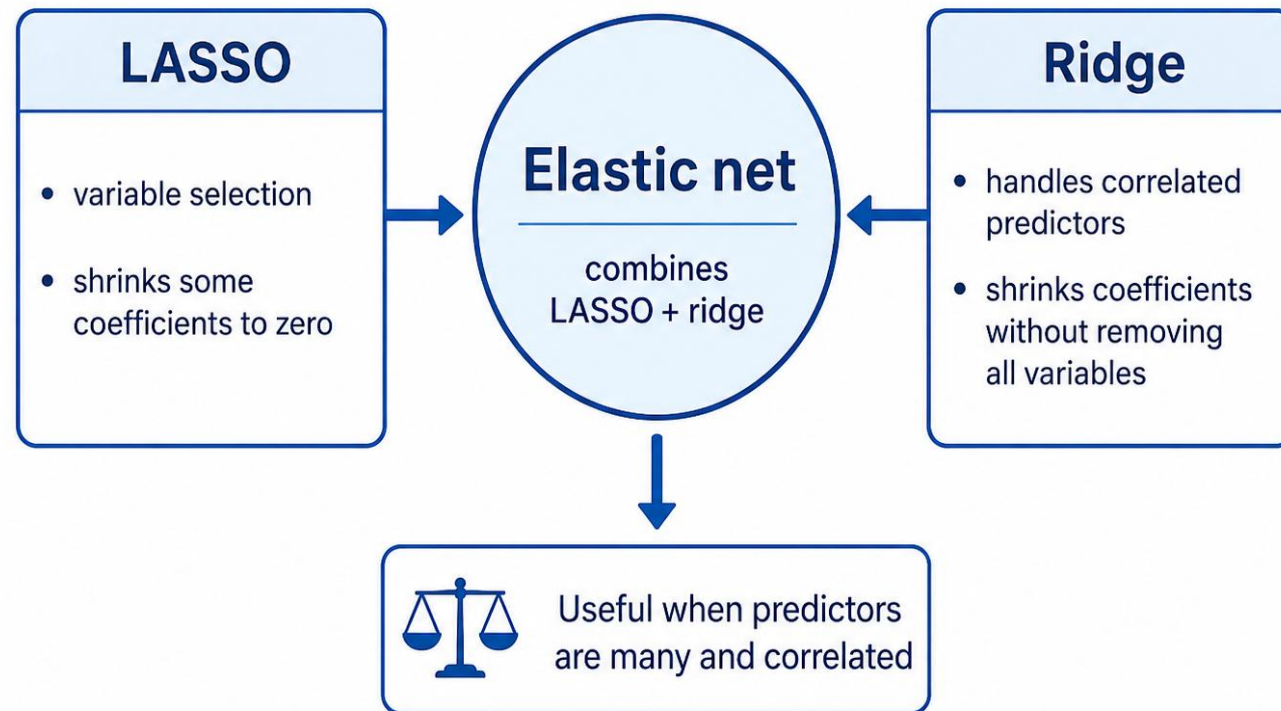
Elastic net regression was used because it can:

- select useful CpG predictors,
- shrink weak effects,
- handle correlated predictors,
- reduce overfitting risk.

$\alpha = 0.5$

a compromise between LASSO and ridge regression

Why elastic net?



How the model was validated

Nested LOOCV

How the model was valid

Nested LOOCV

Outer LOOCV

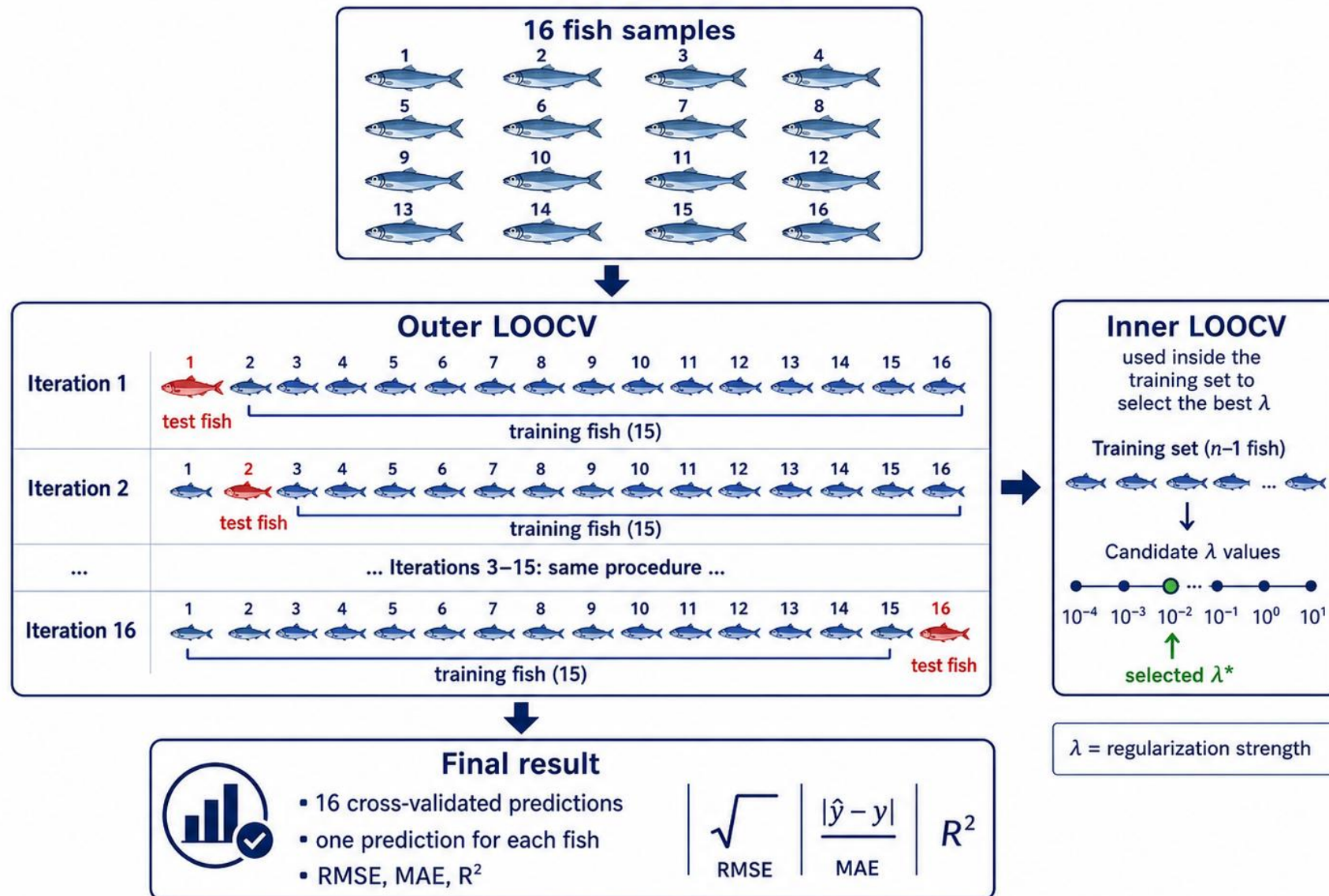
- 1 fish left out for testing
- 15 fish used for training
- repeated 16 times

Inner LOOCV

- used only inside training data
- selects the best λ parameter

Output

- 16 cross-validated age prediction:



Validation changed the picture

Model	RMSE	MAE	R ²
Elastic net without LOOCV	0.07	0.06	0.999
Elastic net with LOOCV	2.16	1.76	0.115



Without validation: almost perfect
With validation: poor prediction

A perfect training fit is not the same as good prediction.

MAE = average error in years

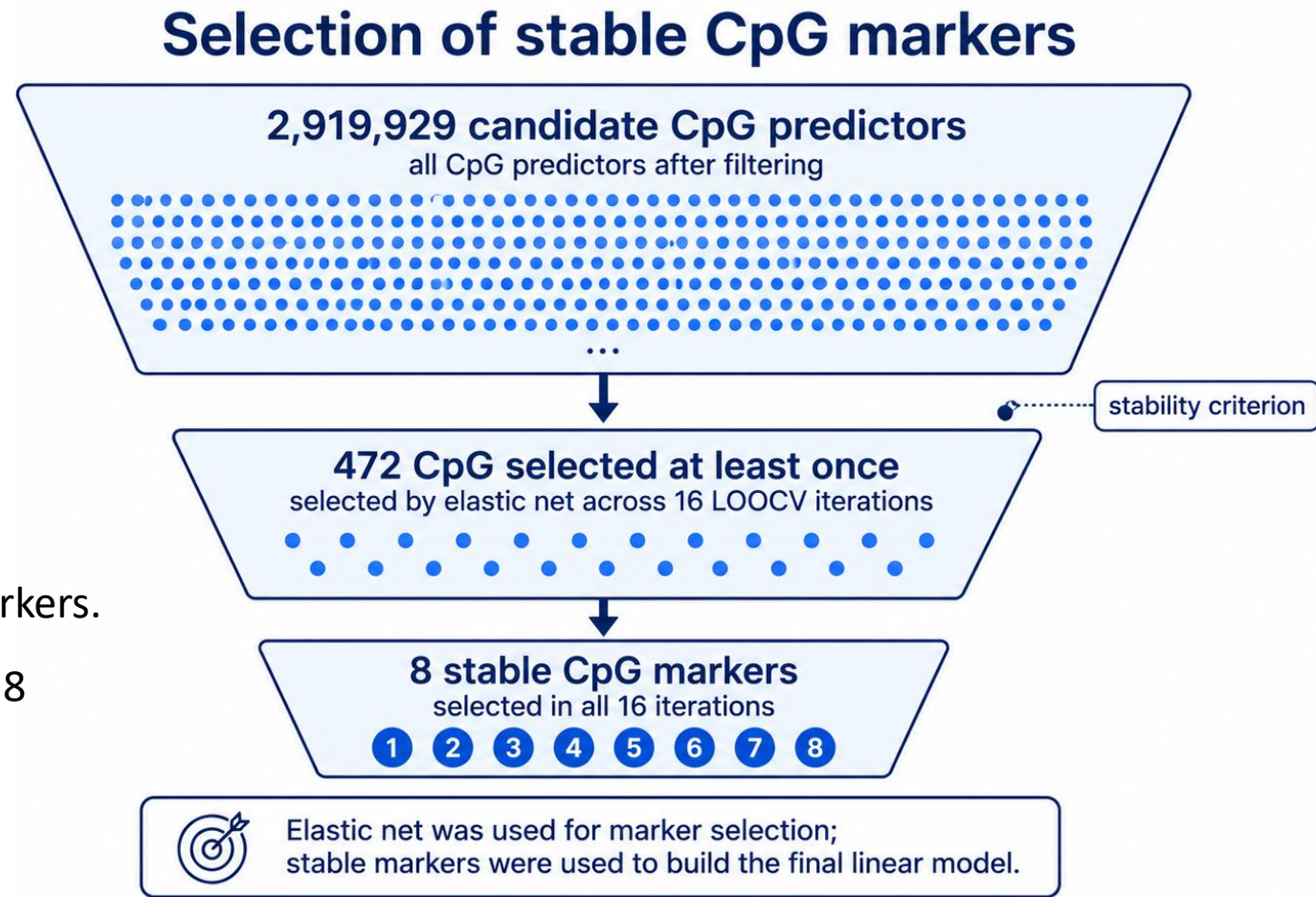
RMSE = error measure sensitive to large mistakes

R² = explained age variation

From elastic net to stable CpG markers

Marker selection across 16 LOOCV iterations

- 472 CpG sites were selected at least once.
- 8 CpG sites were selected in all 16 iterations.
- These 8 sites were treated as stable candidate markers.
- **Next step:** build a simpler model using only these 8 markers.

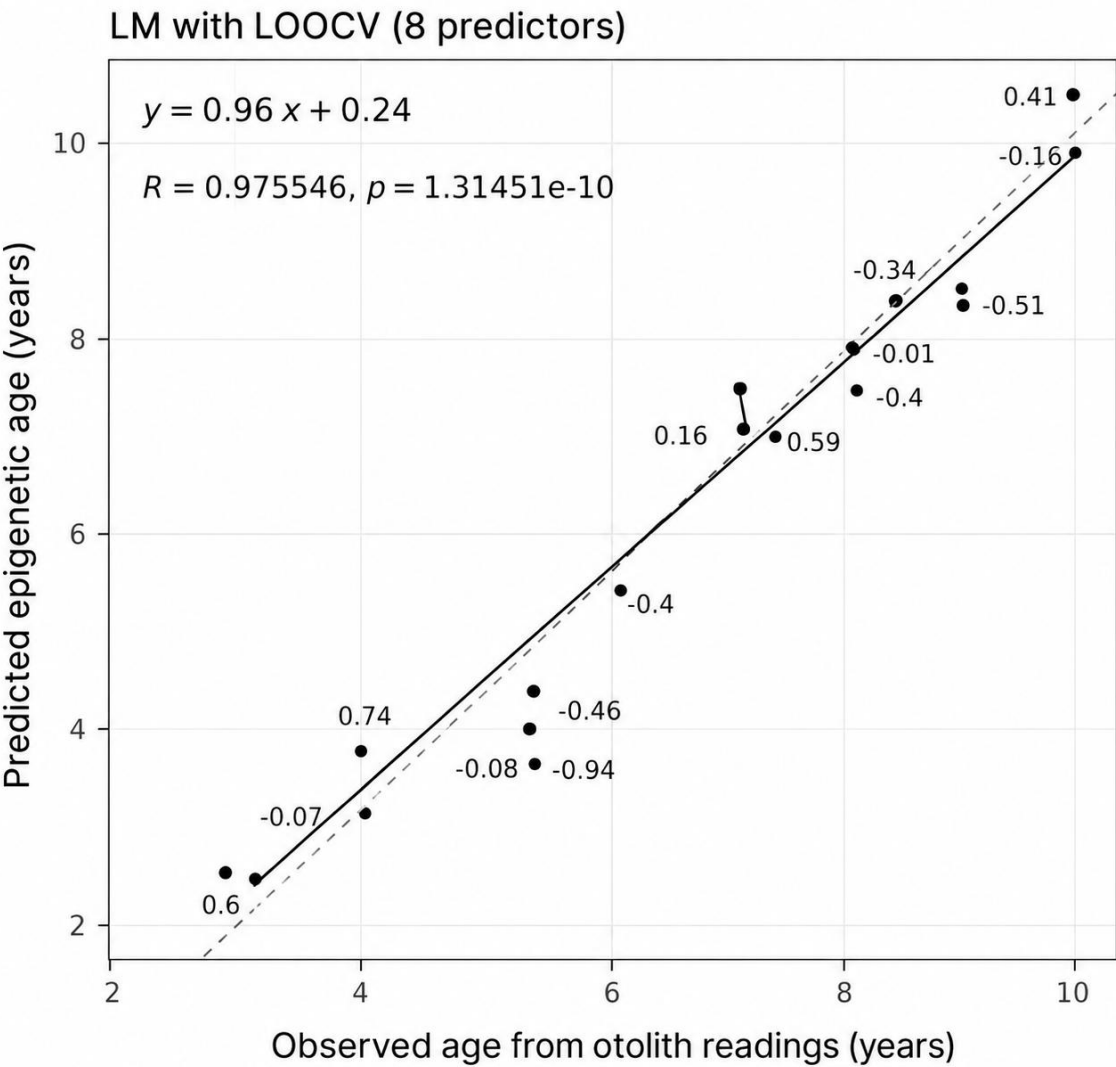


Final pilot model using 8 stable CpG markers

Linear model: age ~ 8 stable CpG markers

Model	RMSE	MAE	R ²
LM without LOOCV	0.23	0.18	0.990
LM with LOOCV	0.50	0.42	0.952

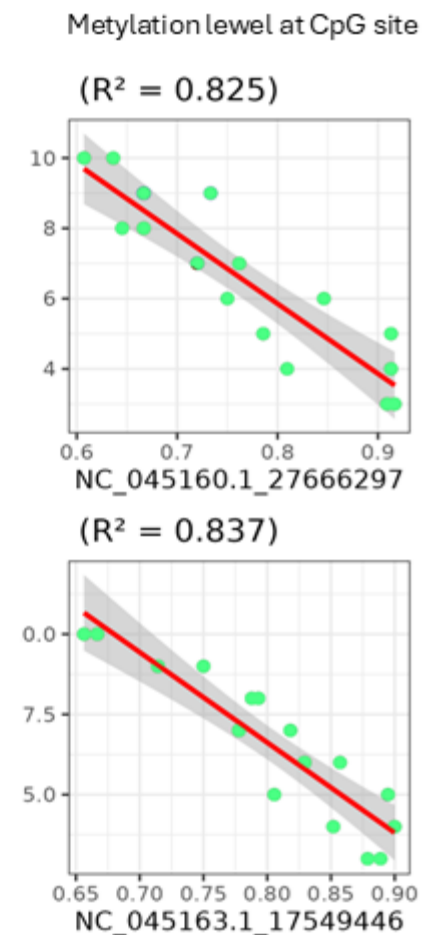
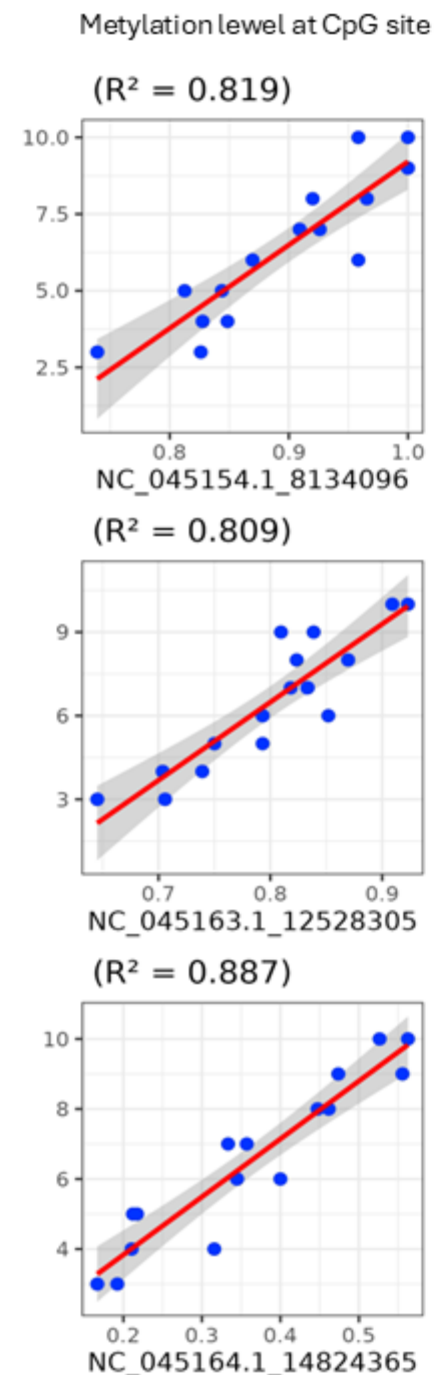
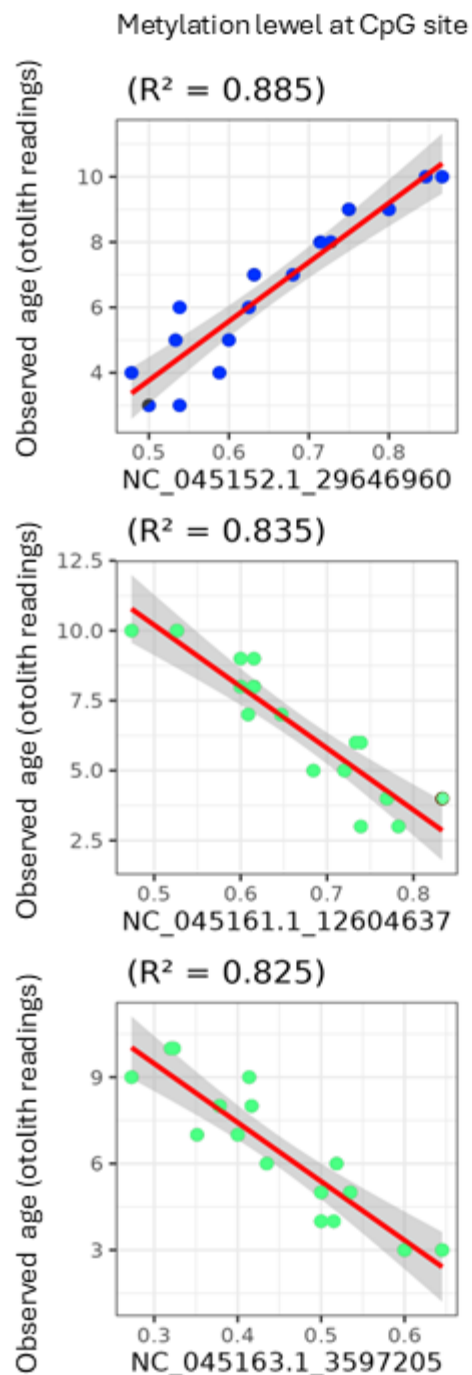
Promising, but still preliminary



Age-related methylation patterns

The 8 selected CpG markers showed strong age-related patterns.

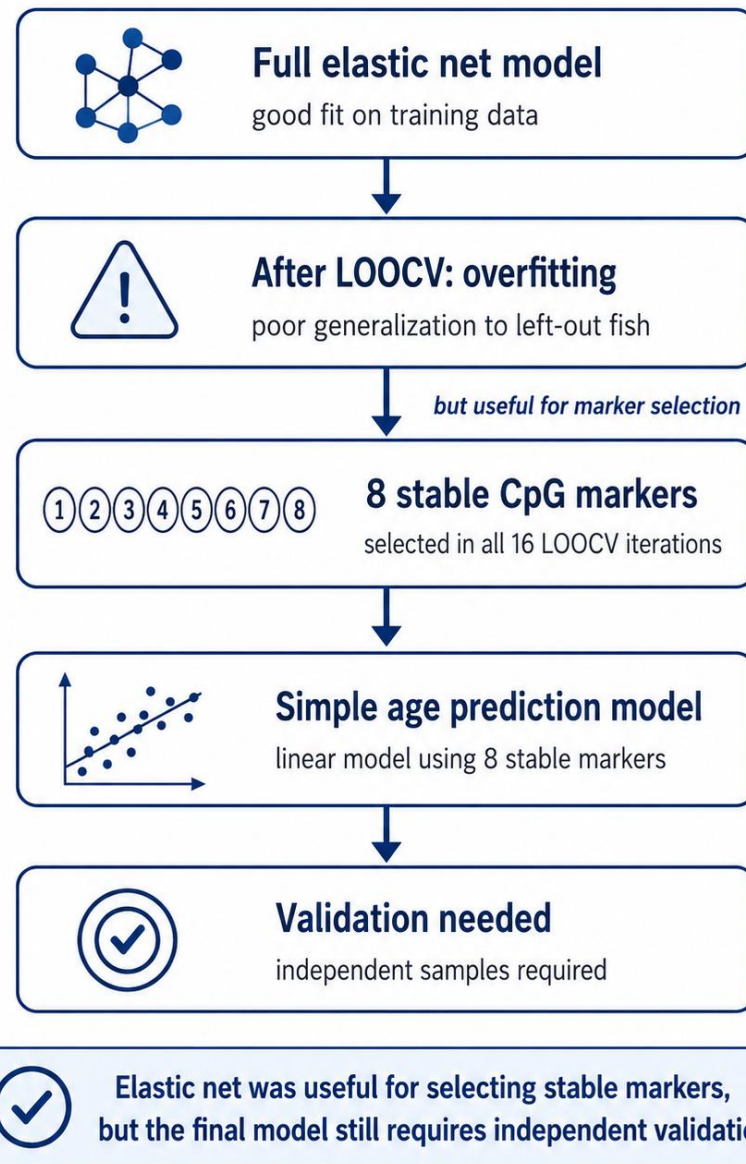
- some markers increased methylation with age,
- some markers decreased methylation with age,
- individual markers were informative,
- the final model combined information from several markers.



Main conclusion from modelling

- The full elastic net model showed signs of overfitting.
- Elastic net was useful for selecting stable CpG markers.
- 8 CpG markers were selected in all LOOCV iterations.
- A simple model using these 8 markers gave promising predictions.
- Independent validation is necessary.

From elastic net to the final age prediction model





Thank you for your attention